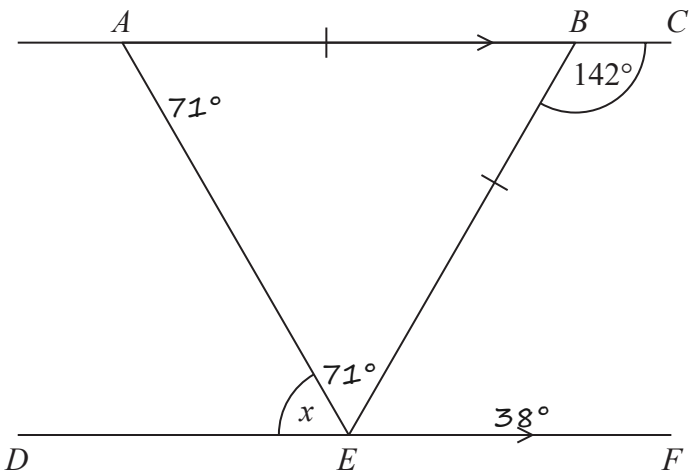
2



ABC and *DEF* are parallel straight lines.
ABE is an isosceles triangle with *AB* = *BE*.
Angle *CBE* = 142°

Work out the size of angle *x*.
Give a reason for each stage in your working.

Angle BEF = 180° – 142° = 38°.

1 mark
Find BEF=38°

Allied angles on parallel lines add to 180°.

1 mark
Allied angles

Angle ABE = 180° – 142° = 38°.

Angles on a straight line add to 180°.

1 mark
Straight line; isosceles; triangle angle sum

AB = BE, so base angles BAE and AEB are equal.

Each is (180° – 38°) ÷ 2 = 71°.

1 mark
Complete angle chain

Angle DEA = angle BAE (alternate angles).

Therefore x = 71°.

1 mark
71°

°

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(Total for Question 2 is 5 marks)